

Personal Ice Resurfacer

Designing a backyard-scale Zamboni for pond hockey and home rinks

Drake Elliott



About Me & Project Background

ABOUT ME



HOMETOWN

Dallas, TX

EDUCATION

MIT '26, B.S. Mechanical Eng.

INDUSTRY

L3Harris OceanServer (AUVs)

RESEARCH

Soft robotics, published & thesis

ATHLETICS

D1 Rowing, Captain 2025–26

PROJECT BACKGROUND



2.009: Product Engineering

MIT's mechanical engineering capstone class

16 person team

6 teams in the class

~3 months to prototype

1 school-wide final demo

Project Objectives & Requirements

DIY RINK MAINTENANCE

COMMERCIAL ZAMBONI

The user need this project targets



PROBLEM

Large gap in the market between **DIY rink maintenance** and **commercial Zambonis**.



TARGET USER

Pond hockey players and **backyard rink owners** in Canada, the Midwest, and the Northeast.



GOAL

Deliver a **commercial-quality, clean surface finish** at a fraction of the cost and scale.

Total System Overview: CONOPS



CONCEPT OF OPERATIONS

- 1 Source water from hose or pond
- 2 Fill onboard tank with hot water
- 3 Drive to resurfacing area
- 4 Lower drum cutter and begin shaving ice
- 5 Deposit heated water onto even surface for clean finish

Total System Overview: Subsystems

01 Chassis and drivetrain



02 Drum cutter



03 Water distribution



My subsystem focus

Focus: My Roles

System Integrator

- Overall architecture
- Coordinate team meetings
- Facilitate discussion
- Cross-subsystem integration

Thermofluid Subsystem Engineer

1 Design

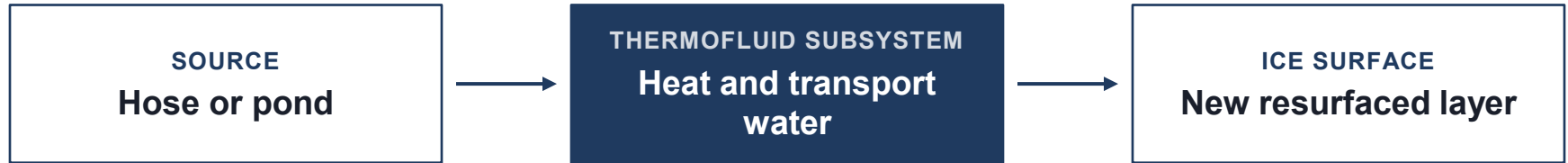
2 Fabrication

3 Testing

4 Implementation



Thermofluid Requirements



100°F

Nominal water temperature
at the ice

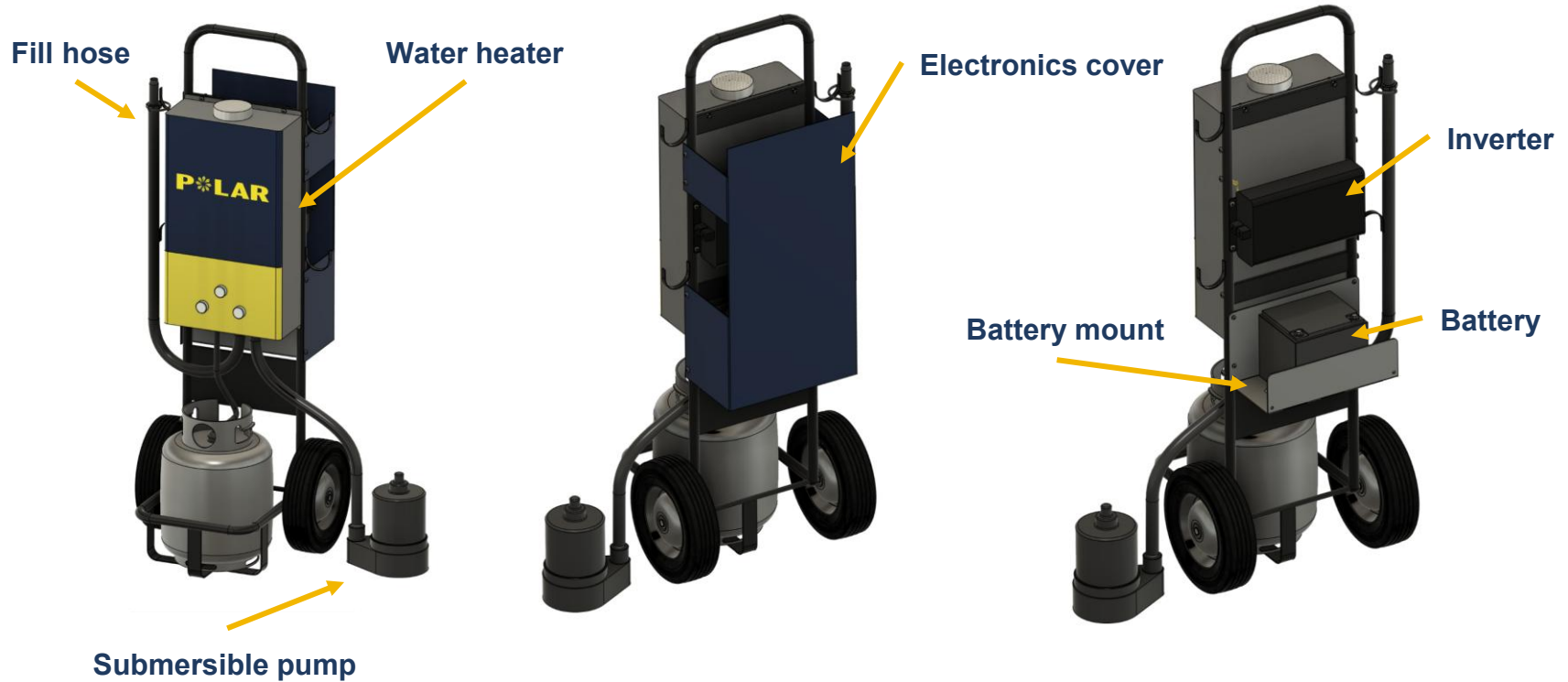
0.025 in

Nominal layer thickness,
held constant

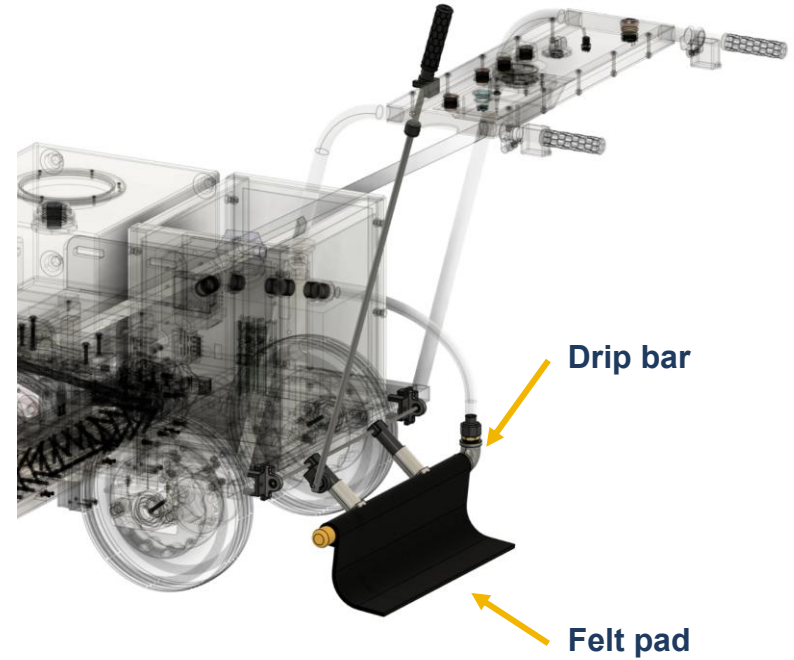
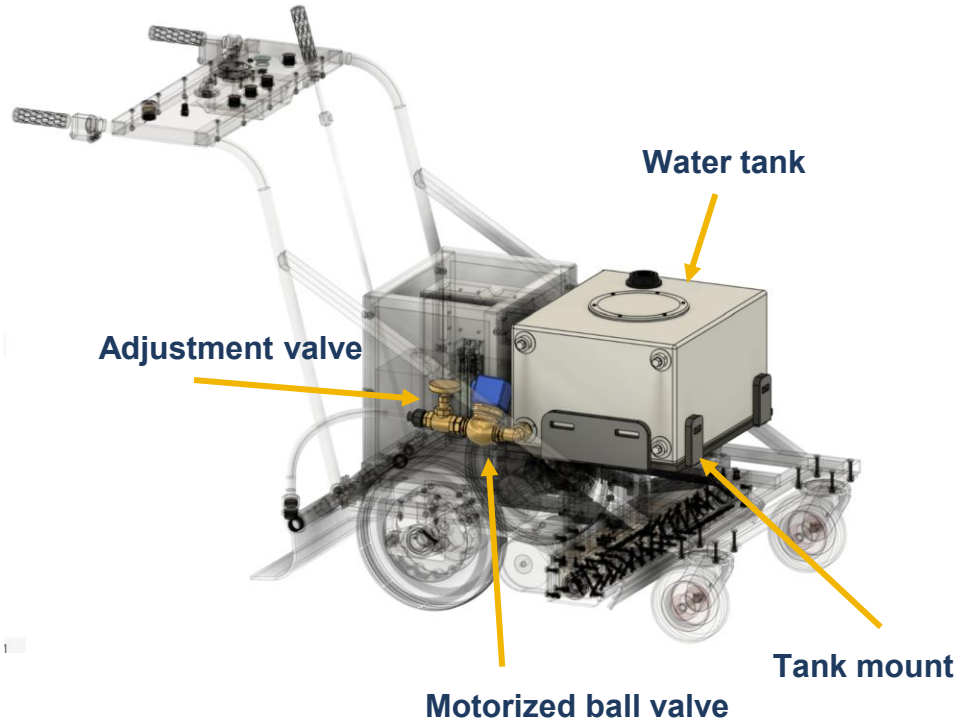
Onboard

Power for the full thermofluid
system

Thermofluid System: Pump-Heater Module



Thermofluid System: Resurfacer



Thermofluid Component Fabrication

01 3D printing



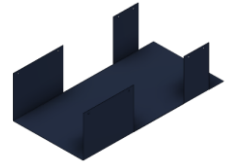
02 Machining



03 Water jetting



04 Sheet metal forming



Challenge 1: Energy Source, Heating Load

Problem: How much energy is required to fulfill the design requirement of heating water? Can this be done electrically?

ASSUMPTIONS

Area covered (A) **2,800 ft²**

Film thickness (t) **0.025 in**

Starting water temp (T_C) **39°F**

Final water temp (T_H) **104°F**

GOVERNING EQUATION

$$E = mc_p\Delta T$$

VALUES

$$\rho_{\text{Water}} = 62.4 \text{ lb/ft}^3$$

$$c_{P, \text{Water}} = 1 \text{ BTU/lb}\cdot^\circ\text{F}$$

RESULT

**~24,000
BTU**

≈ 7 kWh to heat one
2,800 ft² layer

Challenge 1: Energy Source, Battery vs. Propane

Problem: How much energy is required to fulfill the design requirement of heating water? Can this be done electrically?



LiFePO4 Battery



Propane: Selected

Energy density (mass)

90 – 205 Wh/kg

~13,700 – 13,900 Wh/kg

Energy density (volume)

227 – 400 Wh/L

~6,600 – 7,000 Wh/L

Cost density

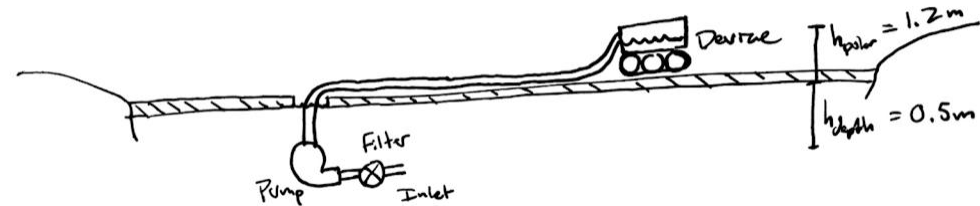
~\$70 – \$150 per kWh

~\$0.09 – \$0.15 per kWh

Challenge 2: Fluid Transport & Pump Specification

Problem: What pump specifications draw water from below the pond and overcome the system's head losses?

SYSTEM SCHEMATIC



GOVERNING EQUATIONS

$$\left(\frac{P}{\rho g} + \frac{v^2}{2g} + z \right)_{\text{inlet}} - \left(\frac{P}{\rho g} + \frac{v^2}{2g} + z \right)_{\text{outlet}} = \sum f \left(\frac{L}{D} \right) \frac{v^2}{2g} + \sum K \frac{v^2}{2g} - h_{\text{pump}}$$

Bernoulli equation

Darcy-Weisbach

Reynolds number

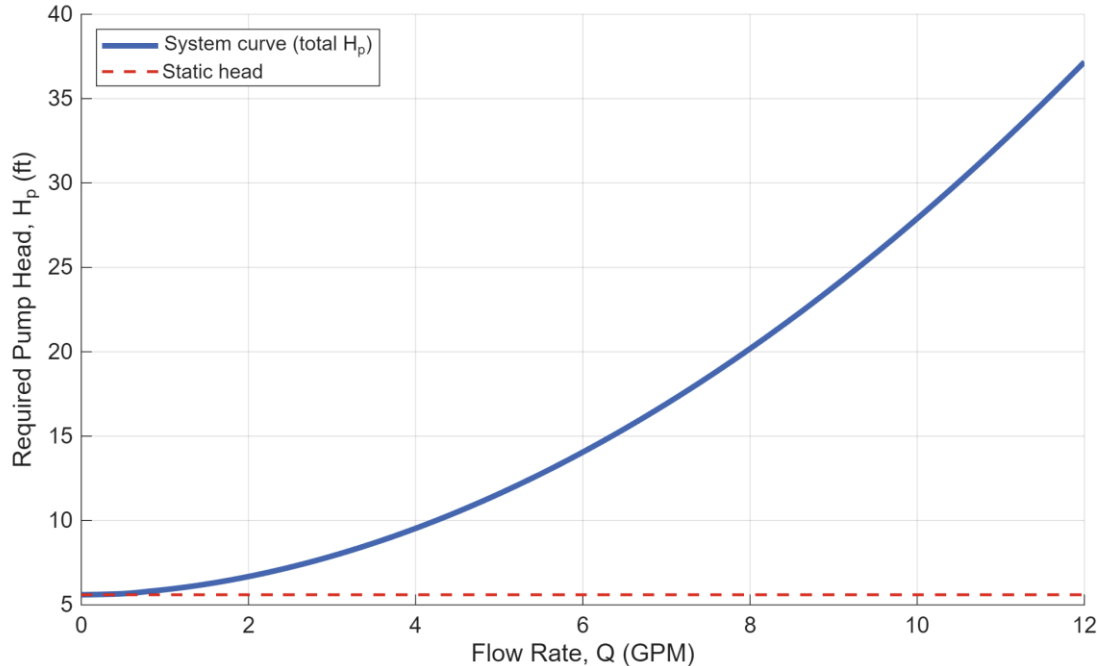
ASSUMPTIONS

Fluid density (ρ)	62.4 lb/ft ³
Kinematic viscosity (ν)	1.93×10 ⁻⁵ ft ² /s
Pipe inner diameter (D)	0.0525 ft
Pipe length (L)	15 ft
Pipe roughness (ϵ)	4.92×10 ⁻⁶ ft
Elevation difference (Δz)	5.6 ft
Total minor loss ($\sum K$)	7.5

Challenge 2: Pump Specification

Problem: What pump specifications draw water from below the pond and overcome the system's head losses?

RESULTS



$Q = 0$ GPM

~5.6 ft

static head only

$Q = 8$ GPM

~20 ft

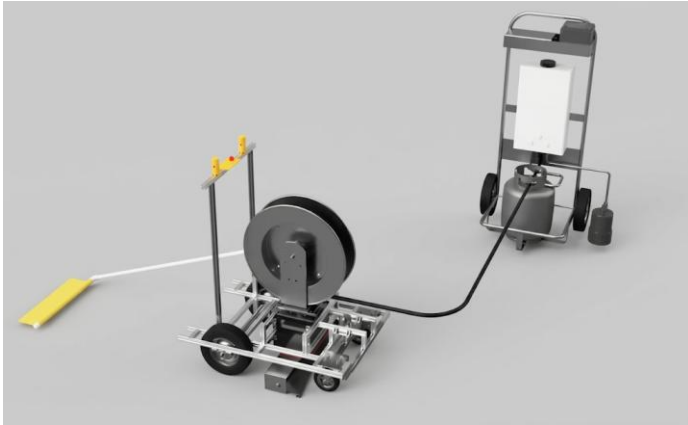
total required head

Challenge 3: Tethered vs. Onboard Tank

Problem: Should the resurfacer be tethered to the pump-heater module, or decoupled with a large onboard tank?

BEFORE

Tethered to pump-heater module



➔
User testing and
feedback

AFTER

Decoupled, large onboard tank

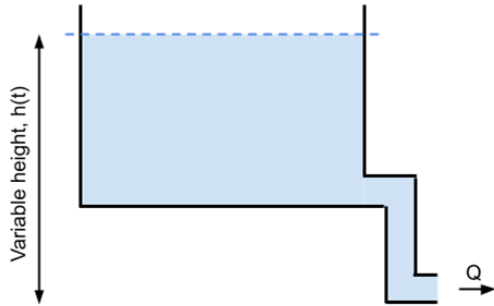


Challenge 3: Constant Flow Without a Pump

Problem: How do we drive a constant flow rate to the drip bar without a pump?

OPTION A

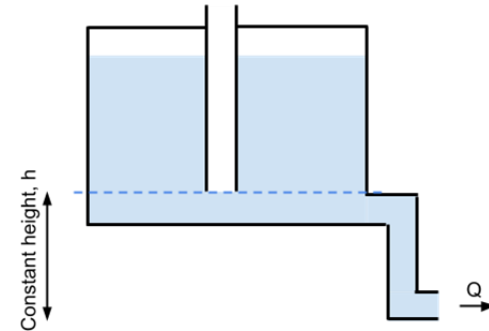
Open tank



$Q = f(t)$ | Higher Q_{avg}

OPTION B: SELECTED

Mariotte tube

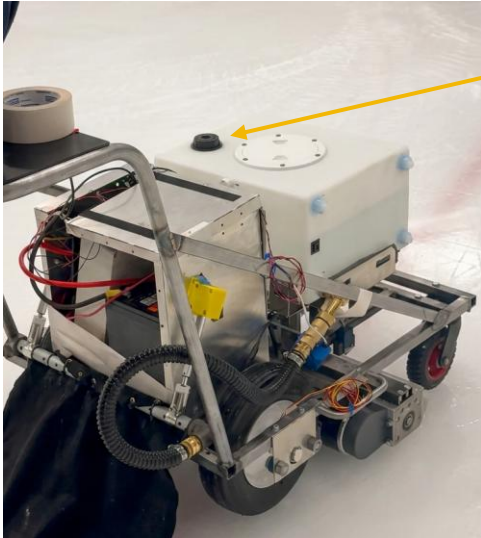


$Q = \text{constant}$ | Lower Q_{avg}

Challenge 3: Mariotte Tube Implementation

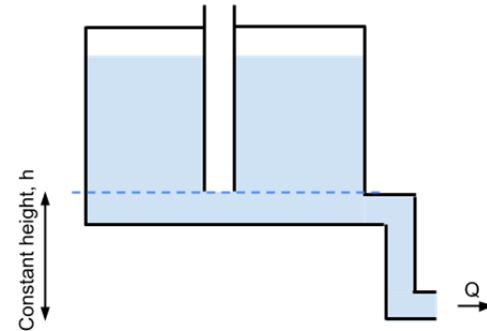
Problem: How do we drive a constant flow rate to the drip bar without a pump?

AS BUILT



PRINCIPLE

Mariotte tube



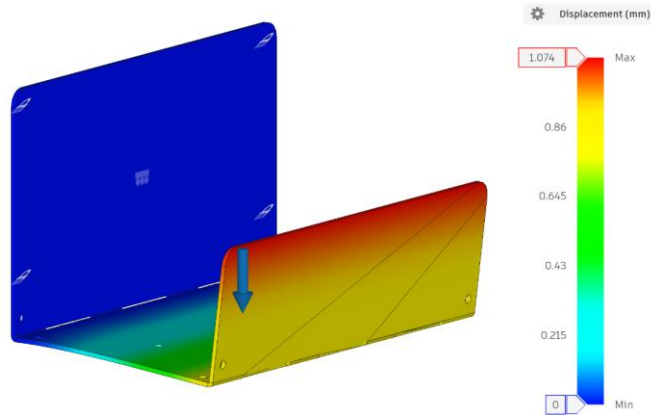
$Q = \text{constant}$

Lower Q_{avg}

Design Validation: Battery Mount Bracket

FEA 1

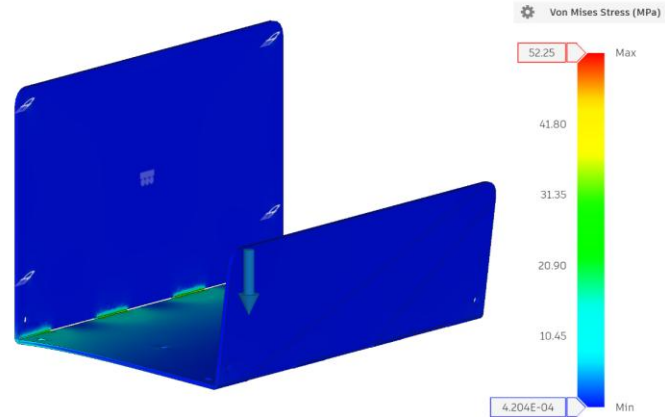
Displacement



$\delta_{\text{Max}} \approx 1 \text{ mm}$
maximum deflection

FEA 2

von Mises stress



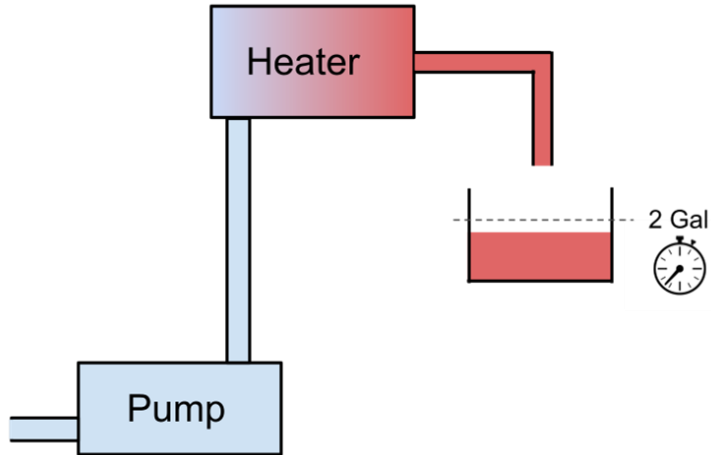
$\sigma_{Y, Al} = 276 \text{ MPa}$
yield

$\sigma_{VM} = 52 \text{ MPa}$
peak stress

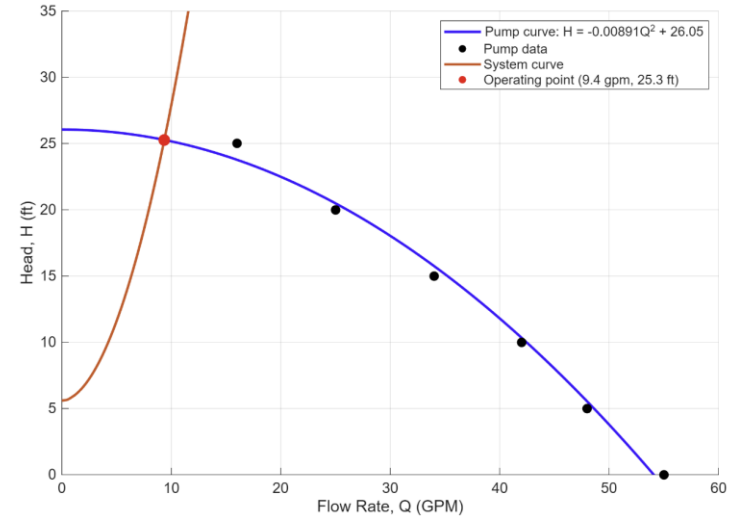
SF = 5.3
safety factor

Design Validation: Fluid Flow

EXPERIMENTAL SETUP



MODEL



25.3 ft

operating point head

9.4 GPM

predicted flow rate

1.5 GPM

experimental result

Outcome & Final Specs

PUMP-HEATER MODULE
FLOW RATE

1.5 GPM

6 min fill time

FLUID TEMPERATURE

104°F

vs. 100°F requirement

WATER DISTRIBUTION

Constant

flow rate and layer
thickness

Final Prototype Demonstration



Full webcast: <https://www.009cycles.com/webcast> @ 2:36:00

Hindsight & Lessons Learned

01

TEST EARLIER

More testing and characterization for the drip bar

02

ORDER EARLIER

Anticipate supplier lead times

03

ASK EARLIER

Query end user before investing in system architecture

Thank You

Questions?

